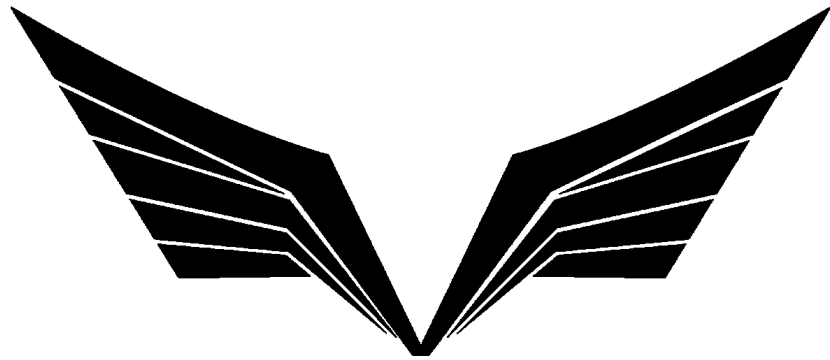




SUNDEVIL

ROCKETRY

Avionics Firmware Development Team

Quality Control Document

Project:	Mod Library
Part Number:	A0015-270
Drafted By:	Eli Sells
Project Engineer:	Eli Sells
Quality Assurance:	Eli Sells
Chief Engineer:	Eli Sells

TABLE OF CONTENTS

1. PROJECT OVERVIEW.....	3
1.1 QUALITY ASSURANCE PRACTICES.....	3
2. SUPPORTING DOCUMENTATION.....	4
3. REQUIREMENTS.....	5
4. DEVELOPMENT READINESS.....	6
5. COMPONENTS.....	7
5.1. SEPARATELY CONTROLLED COMPONENTS.....	7
5.2. COMPONENTS WITH COMBINED CONTROL.....	8
6. TEST PROCEDURES.....	9
7. TEST READINESS.....	10
7.1. CHANGES SINCE PREVIOUSLY REVIEWED BASELINE.....	10
7.2. APPROVAL.....	10
8. TEST RESULTS.....	11
9. FLIGHT QUALIFICATION.....	12
9.1. CHANGES SINCE PREVIOUSLY REVIEWED BASELINE.....	12
9.2. APPROVAL.....	12
10. QUALIFIED VERSIONS.....	13

1. PROJECT OVERVIEW

The mod library is a utility library consisting primarily of platform independent modules written in C that are used throughout SDR's embedded C code.

1.1 QUALITY ASSURANCE PRACTICES

The primary guiding standard for this project is the [SDR Software Quality Assurance Standard](#). This version of the library is intended to be used across all software projects, so it is certified as "Safety Critical".

However, as a library, this project is unable to determine how it is used and is therefore unable to analyze its own hazards. Hazard analysis shall be performed by the integrating project. The architectural guidance for this project is therefore to let error handling be performed by whichever project is calling into it.

The mod library was designed so that no component sits on the system boundary. This therefore mitigates possible security issues.

2. SUPPORTING DOCUMENTATION

This section shall reference any documentation for the controlled article. This could be anything from formal design documents and procedural documentation to informal user's guides associated with the article.

DOCUMENT	DESCRIPTION
Doxygen Documentation	Living documentation from the repository in the _doc folder.

3. REQUIREMENTS

The article's requirements are defined in the _reqs folder under srd.md.

4. DEVELOPMENT READINESS

The project meets the requirements to begin development:

1. Requirements are defined
2. Quality assurance practices have been defined and agreed upon

Date: 9/30/2026

Project Engineer: Eli Sells

Quality Assurance: Eli Sells

5. COMPONENTS

The parent project is required to integrate and verify the “driver” library. No other components are used and no direct dependencies exist.

6. TEST PROCEDURES

All test procedures for this article are defined in code under `_test/mod`. All tests are performed at the unit level. The integrating project is responsible for any integration tests.

7. TEST READINESS

The project has completed its initial development and is ready to enter the testing phase.

7.1. CHANGES SINCE PREVIOUSLY REVIEWED BASELINE

None apply.

7.2. APPROVAL

The project meets the requirements to begin testing:

1. Development is complete and all requirements have been implemented
2. Quality assurance practices have been followed, and an audit has been conducted on a random sample of constituent parts
3. Test procedures have been defined and all requirements will be tested by the procedure

Date: 9/30/2026

Project Engineer: Eli Sells

Quality Assurance: Eli Sells

8. TEST RESULTS

The project has completed its test procedures and have produced the following results:

Requirement tag	Requirement text	Results file	Test status
RQ.MOD.00001	The library shall be designed to interface with the Sun Devil Rocketry "driver" library.	mod/_test/mod/analysis_results.md	PASS
RQ.MOD.00002	The library shall be organized into self-contained modules with header files organized next to source files.	mod/_test/mod/analysis_results.md	PASS
RQ.MOD.00003	The library shall be designed to be compiled with GNU GCC for the desired target architecture.	mod/_test/mod/analysis_results.md	PASS
RQ.MOD.00004	The library shall provide a ping response code based on the provided compiler macro from the project.	mod/_test/mod/commands/results.txt	PASS
RQ.MOD.00005	The library shall provide a method for the flight computer to send a data package to the dashboard over USB.	mod/_test/mod/commands/results.txt	PASS

Requirement tag	Requirement text	Results file	Test status
RQ.MOD.00006	The library shall provide a method for the flight computer to construct the dashboard data packet.	mod/_test/mod/commands/results.txt	PASS
RQ.MOD.00007	The library shall provide a platform-independent method to log debug data.	mod/_test/mod/debug_sdr/results.txt	PASS
RQ.MOD.00008	The debug logger shall use a circular buffer without dynamic memory allocation.	mod/_test/mod/debug_sdr/results.txt	PASS
RQ.MOD.00009	The debug logger shall accept a callback to be used to write debug data asynchronously.	mod/_test/mod/debug_sdr/results.txt	PASS
RQ.MOD.00010	The debug logger shall accept a callback to be used if the buffer overflows.	mod/_test/mod/debug_sdr/results.txt mod/_test/mod/debug_sdr/results.txt	PASS
RQ.MOD.00011	The debug logger shall provide a default implementation of the buffer overflow callback.	mod/_test/mod/debug_sdr/results.txt	PASS
RQ.MOD.00012	The debug logger shall provide a function to be called after an	mod/_test/mod/debug_sdr/results.txt	PASS

Requirement tag	Requirement text	Results file	Test status
	asynchronous write completes.		
RQ.MOD.00013	The library shall provide an error handler.	mod/_test/mod/error_sdr/results.txt	PASS
RQ.MOD.00014	If the USE_ERROR_CALLB ACK macro is defined, the error handler shall search a lookup table for a callback provided by the project given an error code.	mod/_test/mod/error_sdr/results.txt	PASS
RQ.MOD.00015	The error handler shall provide an overridable default error callback if the lookup fails or USE_ERROR_CALLB ACK is not defined.	mod/_test/mod/error_sdr/results.txt	PASS
RQ.MOD.00016	The library shall provide a method to compute a Castagnoli 32-bit cyclic redundancy check.	mod/_test/mod/math_sdr/results.txt	PASS
RQ.MOD.00040	The library shall provide a method to clamp a float within a specified range.	mod/_test/mod/math_sdr/results.txt	PASS
RQ.MOD.00017	Hamilton Product (multiplication)	mod/_test/mod/math_sdr/results.txt	PASS
RQ.MOD.00018	Dot product	mod/_test/mod/math_sdr/results.txt	PASS

Requirement tag	Requirement text	Results file	Test status
RQ.MOD.00019	Addition	mod/_test/mod/math_sdr/results.txt	PASS
RQ.MOD.00020	Scalar Multiplication	mod/_test/mod/math_sdr/results.txt	PASS
RQ.MOD.00021	Unit normalization	mod/_test/mod/math_sdr/results.txt	PASS
RQ.MOD.00022	Conjugation	mod/_test/mod/math_sdr/results.txt	PASS
RQ.MOD.00047	Creation from euler angles	mod/_test/mod/math_sdr/results.txt	PASS
RQ.MOD.00048	Determine whether quat is finite	mod/_test/mod/math_sdr/results.txt	PASS
RQ.MOD.00049	Body-to-world frame transition	mod/_test/mod/math_sdr/results.txt	PASS
RQ.MOD.00050	World-to-body frame transition	mod/_test/mod/math_sdr/results.txt	PASS
RQ.MOD.00041	Addition	mod/_test/mod/math_sdr/results.txt	PASS
RQ.MOD.00042	Cross Product	mod/_test/mod/math_sdr/results.txt	PASS
RQ.MOD.00043	Determine whether vector is finite	mod/_test/mod/math_sdr/results.txt	PASS
RQ.MOD.00044	Determine magnitude	mod/_test/mod/math_sdr/results.txt	PASS
RQ.MOD.00045	Normalization	mod/_test/mod/math_sdr/results.txt	PASS

Requirement tag	Requirement text	Results file	Test status
RQ.MOD.00046	Scalar Multiplication	mod/_test/mod/math_sdr/results.txt	PASS
RQ.MOD.00023	The library shall provide a method to start retrieval of sensor data	mod/_test/mod/sensor/results.txt	PASS
RQ.MOD.00024	The library shall provide a method to retrieve sensor data from the last cycle and start retrieval of sensor data	mod/_test/mod/sensor/results.txt	PASS
RQ.MOD.00025	Dump: Transmit the contents of the last sensor data cycle over USB.	mod/_test/mod/sensor/results.txt	PASS
RQ.MOD.00026	The library shall provide methods to get and set the starting orientation of the flight computer.	mod/_test/mod/sensor/results.txt	PASS
RQ.MOD.00027	Conversion of raw IMU data to floating point values	mod/_test/mod/sensor/results.txt	PASS
RQ.MOD.00028	Integration of acceleration data and body state to yield velocity	mod/_test/mod/sensor/results.txt	PASS
RQ.MOD.00029	Re-mapping raw IMU data based on orientation	mod/_test/mod/sensor/results.txt	PASS

Requirement tag	Requirement text	Results file	Test status
RQ.MOD.00030	Fusion of raw IMU data to yield a unit quaternion orientation estimate	mod/_test/mod/sensor/results.txt	PASS
RQ.MOD.00031	The library shall provide a method to convert a barometric pressure and temperature reading to an ISA altitude.	mod/_test/mod/sensor/results.txt	PASS
RQ.MOD.00032	The library shall provide a method to determine the next message for the flight computer to send.	mod/_test/mod/telemetry/results.txt	PASS
RQ.MOD.00033	A packet containing vehicle position and orientation information	mod/_test/mod/telemetry/results.txt	PASS
RQ.MOD.00034	A packet containing vehicle identification information	mod/_test/mod/telemetry/results.txt	PASS
RQ.MOD.00035	A packet containing vehicle calibration information	mod/_test/mod/telemetry/results.txt	PASS
RQ.MOD.00036	The library shall provide a Mahony filter to fuse accelerometer and gyroscope data to return an attitude estimate.	mod/_test/mod/mahony/results.txt mod/_test/mod/mahony/results.txt mod/_test/mod/mahony/results.txt mod/_test/mod/mahony/results.txt	PASS

DOCUMENT CREATED BY AND INTENDED FOR USE BY SUN DEVIL ROCKETRY Page 16

Requirement tag	Requirement text	Results file	Test status
RQ.MOD.00039	The provided parameters for accelerometer fusion are invalid.	mod/_test/mod/mahony /results.txt mod/_test/mod/mahony /results.txt mod/_test/mod/mahony /results.txt	PASS

9. FLIGHT QUALIFICATION

The project has completed all required testing and has passed with minor or no revisions.

9.1. CHANGES SINCE PREVIOUSLY REVIEWED BASELINE

Updates to automated test procedures to ensure full code coverage. Fixed a bug with the lora preset download feature.

9.2. APPROVAL

The project meets the requirements for flight:

1. Test procedures have been applied and have passed their acceptance criteria
2. A random sample of test procedures shows that they were performed as specified
3. No major design or production revisions have been made since test completion

Date: 10/4/2026

Project Engineer: Eli Sells

Quality Assurance: Eli Sells

Chief Engineer: Eli Sells

10. QUALIFIED VERSIONS

VERSION	QC Status
v2.4.0	Flight Qualified - Grandfathered
v2.5.0	Flight Qualified - Grandfathered
v2.6.0	Flight Qualified - Grandfathered
v2.7.0	Flight Qualified - Safety Critical